

Intuition

let,

$$f_{\vec{w}, b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$$

Now, to avoid overfitting using regularization, we will make w_3 and $w_4 \rightarrow 0$

so, if in cost function

$$\min_{\vec{w}, b} \frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \underbrace{1000 w_3^2 + 1000 w_4^2}_{\Downarrow}$$

If we constraint w_3 and w_4 with such huge numbers then to minimize the overall cost the algorithm will minimize w_3 and w_4
 $\Rightarrow$ Goal Achieved.

Re-writing cost function with Regularization

$$J(\vec{w}, b) =$$

Regularization
 $\Uparrow$ Term

$$\frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \left[\frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right]$$

$\lambda \Rightarrow$ regularization
parameter.

$$+ \left[\frac{\lambda}{2m} b^2 \right]$$

||
can include/
exclude

If λ is too low,
then overfitting occurs

and if λ is too high then
underfitting occurs.